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Multi-instance Multi-label Learning Based on Parallel Attention and Local Label Manifold Correlation

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***Abstract*—Real-world applications always contain complex data objects with multiple semantics. The machine learning tasks generalized from these applications can be formulated as multi-instance multi-label learning (MIML) problems. Each object is represented by a bag of multiple instances and can be associated with multiple labels simultaneously. Extensive studies on the MIML have been conducted during the past few years. However, many of them do not consider local label correlations, which is critical in multi-label scenario. To address this problem, we propose the multi-instance multi-label learning based on parallel attention and local label manifold correlation (MIML-LLMC) algorithm. First, parallel multiple attention mechanisms convert each bag into a number of fusion vectors, one for each label. They are also utilized to find out which instances in the bag trigger respective labels to improve explanability. Second, local label manifold correlation vectors (LLMCs) are constructed based on label manifold generation and clustering. They provide a high level of representation of the label vectors. Third, each fusion vector is concatenated to the LLMC to predict each label independently. With the utilization of attention and LLMCs, MIML-LLMC is able to discover instance-label relations and exploit local label correlations simultaneously. Experiments reveal that our approach is highly competitive to the state-of-the-art MIML algorithms and yields reasonable results in understanding the relations between input patterns and output label semantics.**

***Index Terms*—Attention, Multi-instance multi-label learning, Local label correlation, Label manifold.**

I. Introduction

In traditional supervised learning, each data object is represented by a single instance and associated with a single label. But in most real-world applications, each object can be naturally described by multiple instances and categorized into multiple classes simultaneously. For example, in image classification tasks, an image may have multiple semantics and can be segmented into several regions. Each of them is represented by an instance [1]. Fig. 1 exhibits a classical example. In text classification tasks, the emotional tendencies or described objects of an article may vary in different paragraphs. When a paragraph is extracted as an instance, the tasks naturally become MIML problems [2]. In video detection tasks, multiple objects may appear within one video, and each key frame can be treated as one instance [3]. Hence, the multi-instance multi-label framework is proposed for such complicated tasks [4].

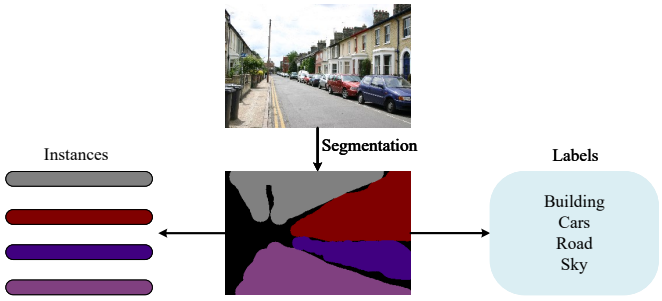


Fig. 1: A classical example of an image with multiple instances and labels. Each region corresponds to an instance. The region of black is discarded.

In recent years, many MIML algorithms have been proposed [1], [5]–[7]. Some of them are developed using the decomposition strategy. MIMLBoost [8] decomposes MIML tasks into a series of multi-instance single-label tasks, and each one corresponds to a label. MIMLSVM [8] decomposes MIML tasks into multi-label tasks by mapping each bag into a single vector. MIMLRBF [9] utilizes RBF kernel for converting bags into vectors and adopts existing multi-label learning algorithms for classification. There are also some works focus on dealing with MIML problems using other learning frameworks [10], [11]. In contrast, some other algorithms aim to tackle MIML problems directly. DMIMLSVM [4] designs a special loss function for the original MIML task. DeepMIML [6] constructs a sub-concept layer and exploits deep neural network formation to generate instance representation for MIML. The main goals of these algorithms can be roughly categorized as: 1) dealing

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with the complicated data consisting of bags; 2) identifying which instance contributes most to which label. Some of the aforementioned algorithms achieve decent performance while investigating the two main goals. However, the label space of MIML is rarely studied.

Exploiting local label correlation has been proven effective in single-instance multi-label learning. As shown in Fig. 2, in



(a) Cow, Grass (b) Swan, Water (c) Bird, Sky

Fig. 2: Illustration of local label correlation.

subfigure 2a, the label "Cow" has a strong correlation with the label "Grass"; in subfigure 2b, the label "Swan" has a strong correlation with the label "Water"; in subfigure 2c, the label "Bird" has a strong correlation with the label "Sky". In contrast, the label "Cow" may not correlate with the label "Water" or "Sky", and vice versa. Such local correlation information can be learned to boost the performance of MIML. The single-instance multi-label algorithm ML-LOC [12] encodes local label correlation into an additional vector for each instance. But since the additional vectors for test data are obtained by a set of linear regressions in ML-LOC, it may result in more uncertainty.

To exploit the information of the label space, we propose the attention based multi-instance multi-label learning with local label correlation in label manifold (MIML-LLMC) algorithm. The attention mechanisms [13] are used to achieve the two goals, i.e., dealing with bag structure and identifying instance contributions. Each attention mechanism allocates a set of weights for instances in a bag and generates a single vector by summing the products between the instances and their corresponding weights. In other words, each bag is converted into multiple vectors, each of which is for the prediction of a specific label. Furthermore, we aim at exploiting label correlation in label manifold to mine useful information from the label space and reduce uncertainty. Traditional multi-label space only contains logical labels, which only describe whether a bag is associated with certain labels. In contrast, the label manifold contains numerical labels that carry more semantic information and can describe the bags more comprehensively [14]. In other words, the label manifold is an extension of logical labels in the Euclidean space. However, the label manifold is not explicitly available in the datasets. We reconstruct the label manifold using a method similar to Local Linear Embedding (LLE) [15]. By constructing a neighborhood graph based on the smoothness assumption [16], the topological structure of bag space is transformed to the label space and the label manifold can be obtained. We further partition the label manifold into several groups using k-Medoids and assume that each group shares a subset of label correlation. The local label manifold correlation vector (LLMC) of each bag is generated as the concatenation of the distance between its numerical label and the centroids. In this way, the LLMCs of test bags can be obtained without other learners. Furthermore, the LLMC of each bag is attached to the vectors converted by attention mechanisms for prediction.

II. Related Work

We briefly review some related works, starting with introducing two learning frameworks related to MIML: multi-instance learning and multi-label learning.

Multi-instance learning (MIL) derives from the drug activity prediction task [17]. In MIL, an object is extracted as a bag consisting of multiple instances and associated with a single label. The labels are only given to the bags while the labels of instances stay unknown. Hence, MIL focuses on dealing with complicated bag structures and thus is rather challenging [18]. MIL has been well studied in recent years and many algorithms have been proposed. The existing MIL methods can be roughly categorized into two groups: instance-based methods and bag-based methods. In the first group, MiFV [19] employs a Gaussian Mixture Model to fit the instance space, and converts each bag into a Fisher vector based on information from each distribution. This enables the MIL issue to be addressed using a single instance classifier. By performing k-Means on the instance space, MiVLAD [19] obtains a single vector of each bag using a difference-based mapping function. StableMI [20] determines the positive information content of an instance in the positive bags, and selects the instances with more positive information to construct the mapping function. In the second group, the simplest way is to take the mean of all instances in the bag as its single vector representation [21]. Bamic [22] uses k-Medoids on the bag-level and generates instance representation for each bag. Each feature of the representation is the distance between the bag and each centroid. ELDB [23] selects more representative bags through the discriminative analysis and reinforcement technique, and finally obtains more distinguishable single vectors. Attention-net [24] designs a two-layered network to assign weights for the instances in each bag. Loss-attention [25] fuses loss functions and attention mechanisms to make instance-level predictions.

Multi-label learning (MLL) studies the problem where each object is represented by a single instance while associated with a set of labels simultaneously. Some MLL algorithms work in a decomposition way. A classical method is to break down a MLL task into a series of single-label tasks, and each label is learned independently [26]. Others tend to explore label correlation, which exists naturally in the multi-label space. ML-LOC [12] encodes the local label correlation into additional features for each instance. Following that, Zhu and et al. propose GLOCAL algorithm to capture global and local label correlation simultaneously [27]. In MLL, the issue of missing labels is quite relevant, Cheng and et al. employ an attention mechanism to obtain a high quality recovery of missing labels [28].

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MIL is dedicated to addressing the issue that the features of one object cannot be simply concatenated for developing machine learning algorithms. MLL is concerned with the issue of an object belonging to many categories at once. In real-world applications, these two issues are naturally existing and fairly prevalent. To tackle the issues once and for all, the framework of MIML is formalized [4], where an object is represented by multiple instances and associated with multiple labels.

In recent years, many algorithms have been proposed to address MIML problems. Some of them focus on traditional ways. By individually selecting representative bags within ones belonging to each class, MIMLRBF [9] directly exploits the connections between instances and labels. MIML-kNN [11] employs a k-nearest algorithm that not only considers one bag's neighbors, but also considers its citers which regard it as their own neighbors. To discover which instance triggers which label, KISAR [5] generates a prototype for each label based on clustering. Briggs et al. [7] proposes a label ranking loss to address a MIML bird species recognition problem. To handle large scale data at a feasible time cost, MIMLFast [1] defines a low-dimensional subspace shared by all labels and an optimization method with randomly sampled triples. There are a few works attempt to apply MIML to deep neural networks. DeepMIML [6] constructs a sub-concept layer to automatically locate the key input patterns that trigger the labels, thus offering a solution to deep MLL in a MIML way. MIML-FCN+ [29] designs a two-stream fully convolutional network and optimizes it under the guidance of the privileged information loss. Some works tend to focus on practical issues with the MIML framework, such as action recognition [30], video caption [31] and surgical image annotation [32]. The aforementioned algorithms demonstrate the usefulness of the MIML framework. Hence, some works try to extend the MIML framework to novel settings. Xing et al. [33] propose a way to deal with multi-view multi-instance multi-label problems using collaborative matrix factorization. MVMIMLA [34] proposes to achieve the same goal with the involvation of active learning.

III. The MIML-LLMC Approach

Let $\mathcal{X} = \{\boldsymbol{X_i}\}_{i=1}^{N}$ be the set of N bags and $\mathcal{Y} = \{\boldsymbol{Y}_i\}_{i=1}^{N}$ be the respective labels vector set. $\boldsymbol{X}_i = [\boldsymbol{x}_{i1}, \boldsymbol{x}_{i2}, \ldots, \boldsymbol{x}_{in_i}]$ is the i-th bags, where $\boldsymbol{x}_{ij}$ is the j-th instance and n_i is the bag size. $\boldsymbol{Y}_i = [y_{i1}, y_{i2}, \ldots, y_{iL}]$ is the label vector of i-th bag, where L is the number of labels; if the bag is associated with j-th label, $y_{ij} = 1$, otherwise $y_{ij} = 0$. We use $\mathcal{M} = \{\boldsymbol{M}_i\}_{i=1}^{N}$ represents the label manifold, where $\boldsymbol{M}_i = [m_{i1}, m_{i2}, \ldots, m_{iL}]$ is the numerical label vector of i-th bag. In the rest of this section, we will first introduce the reconstruction process of the label manifold, and present the method to generate the local label manifold correlation vectors (LLMCs). After that, the prediction model is illustrated.

A. Label Manifold

According to the smoothness assumption, the topological structure of the bag space can be transferred to the label space local by local. In order to keep the locality, we construct a neighborhood graph $\mathcal{G} = \langle \mathcal{V}, \mathcal{E}, \boldsymbol{W} \rangle$, where $\mathcal{V} = \mathcal{X}$ is the vertex set in which each bag is a vertex, $\mathcal{E}$ is the edge set in which each edge e_{ij} represents the relationship between the bag $\boldsymbol{X}_i$ and $\boldsymbol{X}_j$, and $\boldsymbol{W}$ is the weight matrix in which each element w_{ij} is the weight of the edge e_{ij}. The value of w_{ij} is defined as:

$$w_{ij} = \begin{cases} \dfrac{1}{D(\boldsymbol{X}_i, \boldsymbol{X}_j)}, & \text{if } \boldsymbol{X}_j \in \mathcal{N}_i; \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where $\mathcal{N}_i$ is the k-nearest neighbors of $\boldsymbol{X}_i$ and $|\mathcal{N}_i| = K$, $D(\boldsymbol{X}_i, \boldsymbol{X}_j)$ is the average Hausdorff distance:

$$D(\boldsymbol{X}_i, \boldsymbol{X}_j) = \frac{\sum\limits_{\boldsymbol{x}_{ia}} \min\limits_{\boldsymbol{x}_{jb}} d(\boldsymbol{x}_{ia}, \boldsymbol{x}_{jb}) + \sum\limits_{\boldsymbol{x}_{jb}} \min\limits_{\boldsymbol{x}_{ia}} d(\boldsymbol{x}_{jb}, \boldsymbol{x}_{ia})}{n_i + n_j}, \tag{2}$$

in which $d(\boldsymbol{x}_{ia}, \boldsymbol{x}_{jb})$ is the Euclidean distance. According to Eq. (1), the edge weight between the bags that are closer should be higher. In this way, the weight matrix $\boldsymbol{W}$ simply represents the local topological structure of the bag space. Once $\boldsymbol{W}$ is obtained, the softmax function is applied to the edge weights of each bag $\boldsymbol{X}_j \in \mathcal{N}_i$:

$$w_{ij} = \frac{\exp(w_{ij})}{\sum\limits_{\boldsymbol{X}_k \in \mathcal{N}_i} \exp(w_{ik})}. \tag{3}$$

To transfer the locality from bag space to the label space, we utilize $\boldsymbol{W}$ to reconstruct the numerical label $\boldsymbol{M}_i = \{m_{i1}, m_{i2}, \ldots, m_{iL}\}$ for bag $\boldsymbol{X}_i$, the element m_{ij} is computed as:

$$m_{ij} = \sum_{\boldsymbol{X}_k \in \mathcal{N}_i} w_{ik} \cdot y_{kj}. \tag{4}$$

The softmax function is also needed here to convert all elements in $\boldsymbol{M}_i$ into probabilities:

$$m_{ij} = \frac{\exp(m_{ij})}{\sum\limits_{m_{ik} \in \boldsymbol{M}_i} \exp(m_{ik})}. \tag{5}$$

So far, the label manifold $\mathcal{M}$ is obtained.

B. Local Label Manifold Correlation

Inspired by [12], the local label manifold correlation for each bag is encoded as an additional vector (LLMC). Once the label manifold $\mathcal{M}$ is acquired, a k-Medoids is performed to partition it into several groups i.e., $\mathcal{M} = \cup\{\boldsymbol{G}_i\}_{i=1}^{R}$ where R is the number of clusters and if $i \neq j$, $\boldsymbol{G}_i \cup \boldsymbol{G}_j = \emptyset$. We assume that the bags whose numerical labels lie in same group share the same local label correlation. For each group $\boldsymbol{G}_i$, we denote its centroid $\boldsymbol{g}_i$ as a prototype to represent its internal numerical labels. Furthermore, the LLMC of $\boldsymbol{X}_i$ is represented by $\boldsymbol{c}_i = [c_{i1}, c_{i2}, \ldots, c_{iR}]$, the element c_{ij} is computed as:

$$c_{ij} = d(\boldsymbol{M}_i, \boldsymbol{g}_j), \tag{6}$$

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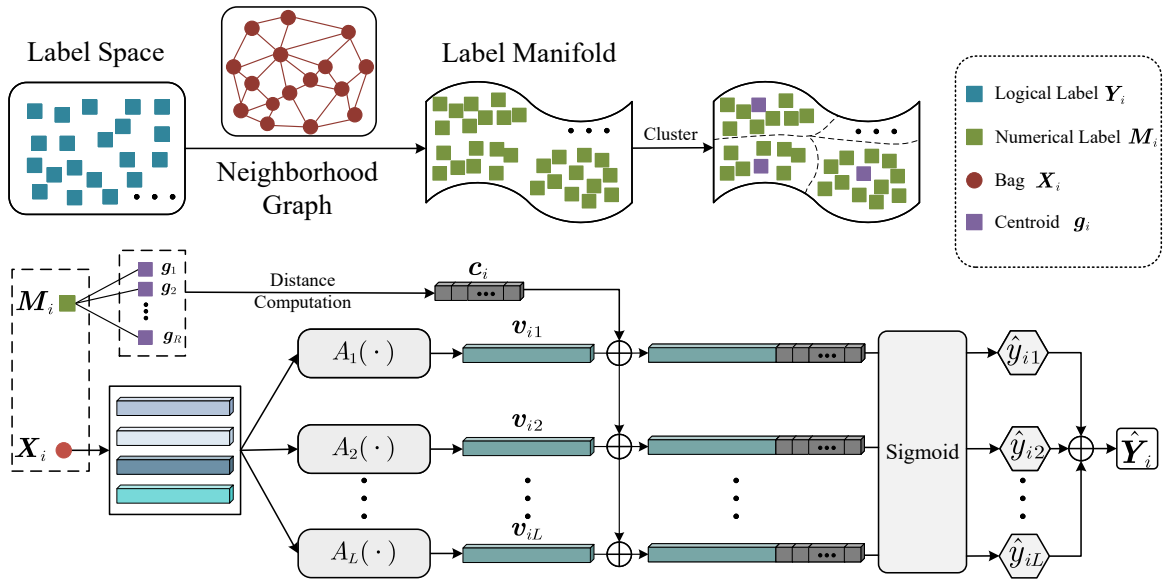


Fig. 3: The framework of the proposed model. It is worth noting that each branch of our model takes into account local label correlation. $A_i(\cdot)$ denotes the attention mechanism which will be introduced later in detail, and $\oplus$ refers to the concatenation between vectors or values. $\hat{\boldsymbol{Y}}_i$ denotes the predicted label vector of $\boldsymbol{X}_i$.

where $d(\boldsymbol{M}_i, \boldsymbol{g}_j)$ represents the Euclidean distance. The value of c_{ij} suggests the correlation between $\boldsymbol{M}_i$ and all numerical labels in $\boldsymbol{G}_i$. Hence, the local label correlation is encoded in $\{\boldsymbol{c}_i\}_{i=1}^N$, which is helpful to obtain a better performance.

C. Prediction Model

Previous works [24], [25] have demonstrated the effectiveness of using attention mechanisms to deal with bag structure. In this paper, we construct a attention based network with multiple branches, each branch predicts one label. By assigning an attention mechanism to each label individually, our model can handle the bag structures and learn instance-label relations at the same time. Our network is designed based on decomposition strategy, some other works based on the similar strategy have achieve adequate performance [1], [4], [8]. However, they all focus on exploring instance-label relations while ignoring the investigation of label space. It may result in unsatisfactory performance. Hence, the LLMCs are embedded into the network to exploit the local label correlation. The framework of the proposed model is shown in Fig. 3. We employ a set of attention mechanisms $\{A_1(\cdot), A_2(\cdot), \ldots, A_L(\cdot)\}$ to generate a matrix $\boldsymbol{V}_i = [\boldsymbol{v}_{i1}, \boldsymbol{v}_{i2}, \ldots, \boldsymbol{v}_{iL}]$ for bag $\boldsymbol{X}_i$, i.e., $A_j(\boldsymbol{X}_i) = \boldsymbol{v}_{ij}$.The LLMC $\boldsymbol{c}_i$ of bag $\boldsymbol{X}_i$ is attached to each fusion vector $\boldsymbol{v}_{ij}$ in $\boldsymbol{V}_i$ for prediction.

As aforementioned, the attention mechanism can assign appropriate weights to each instance in one bag and aggregate the bag into a fusion vector. Fig. 4 illustrates the calculation process of the attention weights.

In detail, $P_j(\cdot)$ serves as a transformation to obtain a sufficient expressive power of bag $\boldsymbol{X}_i$ by mapping it into a new feature space. With the output of $P_j(\cdot)$, $Q_j(\cdot)$ is utilized to learn the attention weights for the k-th instance:

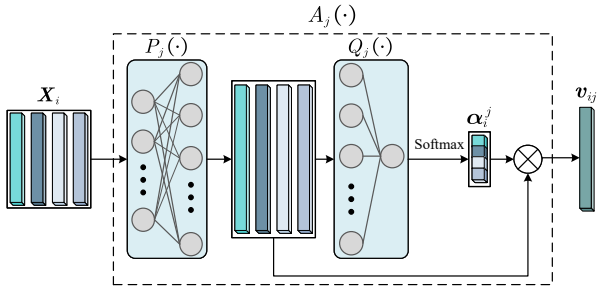


Fig. 4: An illustration of the j-th attention mechanism. The given bag $\boldsymbol{X}_i$ is converted into a single vector by summing the products between each instance $\boldsymbol{x}_{ik}$ and its corresponding weight α_{ik}^j. $P_j(\cdot)$ is a transformation and $Q_j(\cdot)$ is a single layer network with one output unit. $\otimes$ represents the dot production.

$$\alpha_{ik}^j = Q_j(P_j(\boldsymbol{x}_{ik})). \tag{7}$$

For comparison, a softmax function is employed to normalize attention weights of $A_j(\cdot)$:

$$\alpha_{ik}^j = \frac{\exp(\alpha_{ik}^j)}{\sum_{t=1}^{n_i} \exp(\alpha_{it}^j)}. \tag{8}$$

Simultaneously, according to Eqs. (7) and (8), an activation function LeakyReLU is used to fit nonlinear information.

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Therefore, the entire calculation process of the attention weight α_{ik}^{j} can be formulated as:

$$\alpha_{ik}^{j} = \frac{\exp(\text{LeakyReLU}(Q_j(P_j(\boldsymbol{x}_{ik}))))}{\sum_{t=1}^{n_i} \exp(\text{LeakyReLU}(Q_j(P_j(\boldsymbol{x}_{it}))))}, \tag{9}$$

in which α_{ik}^{j} reveals the contribution of the k-th instance in the i-th bag to the j-th label. Once the weights $[\alpha_{i1}^{j}, \alpha_{i2}^{j}, \ldots, \alpha_{in_i}^{j}]$ for all instances in bag $\boldsymbol{X}_i$ is obtained, the fusion vector can be computed as:

$$\boldsymbol{v}_{ij} = \sum_{k=1}^{n_i} \alpha_{ik}^{j} \cdot P_j(\boldsymbol{x}_{ik}). \tag{10}$$

Furthermore, we compute the local label manifold correlation vector (LLMC) $\boldsymbol{c}_i$ of bag $\boldsymbol{X}_i$ according to Eq. (6). And it is attached to the fusion vector $\boldsymbol{v}_{ij}$ to obtain $\hat{y}_{ij}$ individually. Let $F_j(\cdot)$ denote a network which have one output unit and the Sigmoid function scales the output data between 0 and 1, the predicted label $\hat{y}_{ij}$ is expressed as:

$$\hat{y}_{ij} = \text{Sigmoid}(F_j(\boldsymbol{v}_{ij} \| \boldsymbol{c}_i)), \tag{11}$$

where $\|$ represents the concatenation operation. Notably, all the above transformations and network layers employ bias but the statement is omitted for brevity.

D. Key Instance Localization

In most real-word applications, only the objects are labeled, while the labels of semantics are not available. For example, in image classification tasks, labels are only given to whole images because the annotations of inside semantics require a lot of time and effort. Due to the explanability provided by the parallel attention mechanisms, the proposed algorithm can identify the key instance that triggers each label. Notably, we follow the assumption that there is only one label which can be triggered by one key instance [1]. Once the weights generated by all attention mechanisms are obtained, we assume the instance $\boldsymbol{x}_{ik}$ triggers ξ-th label , it can be computed as:

$$\xi = \arg\max_{j \in [1,\ldots,L]} \alpha_{ik}^{j}. \tag{12}$$

For the data which do have instance labels, we propose a way to learn both bag labels and instance labels simultaneously. Inspired by [25], the attention mechanisms are directly connected to an instance-level loss function. In detail, a softmax function converts attention weights of $\boldsymbol{x}_{ik}$ into probabilities:

$$\hat{y}_{ik}^{j} = \frac{\exp(\alpha_{ik}^{j})}{\sum_{t=1}^{L} \exp(\alpha_{ik}^{t})}, \tag{13}$$

where $\hat{y}_{ik}^{j}$ is the probability that $\boldsymbol{x}_{ik}$ belongs to label y_j.

Let $\hat{\boldsymbol{Y}}_i = [\hat{y}_{i1}, \hat{y}_{i2}, \ldots, \hat{y}_{iL}]$ be the predicted label vector of bag $\boldsymbol{B}_i$, $\hat{\boldsymbol{Y}}_{ij} = [\hat{y}_{ij}^{1}, \hat{y}_{ij}^{2}, \ldots, \hat{y}_{ij}^{L}]$ be that of instance $\boldsymbol{x}_{ij}$. The bag-level loss function is denoted by $\mathcal{L}^{B}()$ and instance-level loss function is $\mathcal{L}^{I}()$. The overall loss can be formed as:

$$\mathcal{L} = \sum_{i=1}^{n} \mathcal{L}^{B}(\hat{\boldsymbol{Y}}_i, \boldsymbol{Y}_i) + \lambda \sum_{i=1}^{n} \sum_{j=1}^{n_i} \mathcal{L}^{I}(\hat{\boldsymbol{Y}}_{ij}, \boldsymbol{Y}_{ij}), \tag{14}$$

where λ is the trading-off parameter.

IV. Experiments

A. Evaluation

MIML tasks have more complicated label structure thus the metrics of traditional supervised learning is not applicable. In our experiments[1], we employ 5 commonly used MIML metrics [4]: $Hamming\ Loss$, $One\ Error$, $Coverage$, $Ranking\ Loss$ and $Average\ Precision$. For the first 4 metrics, a smaller value indicates better performance, while for $Average\ Precision$, the larger, the better. Notable, the value of $Coverage$ is divided by the number of categories so that the values of all metrics are in the interval $[0, 1]$. In addition, each result is obtained by 10 times 3-fold cross validation.

B. Datasets

Our experiments are conducted on two groups of datasets. Based on the number of categories they obtain, we call them moderate-classed ($L \ll N$) datasets and large-classed datasets.

Among the moderate-classed datasets, Scene and Reuters are collected by Zhou et al. and are the mostly used datasets in existing MIML works. Scene contains 2000 bags, each one is extracted from a scene image by SBN [35] and may be associated with 5 labels: desert, mountains, sea, sunset, and trees. Reuters is derived from the REUTERS-21578 dataset and only the 7 most frequent categories are considered. Each document is extracted as a bag of instances using the method proposed in [36]. The other 3 datasets are collected by Fern et al. in [7]. Letter Carroll and Letter Frost are generated using two poems, "Jabber-wocky" and "The Road Not Taken" from the UCL Letter Recognition dataset [37]. Each word in these poems corresponds to one bag. MSRC v2 is the latest version of the image dataset proposed by Microsoft Research Cambridge (MSRC). Each image comes with its ground-truth segmentation, so that histograms of gradients and colors are extracted to form an instance for each region in the segmentation.

The large-classed datasets are collected by Wu et al. in [38]. We utilize the datasets from the domain of archaea and bacteria genomes, each genomes is extracted into two datasets. The two datasets derived from archaea genomes Haloarcula marismortui and Pyrococcus furiosus are denoted by HAL and PYR, while the other ones from bacteria genomes Geobacter sulfurreducens and Azotobacter vinelandii are denoted by GEO and AZO. The number of classes for these datasets is much higher and can even be comparable to the number of bags. Table I exhibits some properties of the datasets, they are sorted by the number of labels from largest to smallest.

[1]the source code of MIML-LLMC and some datasets are available at https://github.com/heivens/MIML-LLMC.

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TABLE I: Properties of the datasets.

Dataset	Label	Bag	Instance	Label per Bag
Scene	5	2000	18000	1.24
Reuters	7	2000	7119	1.15
MSRC v2	23	591	1758	2.51
Letter Carroll	26	166	717	3.93
Letter Frost	26	144	565	3.60
HAL	234	304	950	3.25
GEO	320	379	1214	3.14
PYR	321	425	1317	4.48
AZO	340	407	1251	4.00

C. Parameter Analysis

The main parameters in our algorithm are the number of neighbors K and the number of clusters R. The value of R is set to dynamically change with the number of labels L in the dataset, i.e., $R = \lfloor p \cdot L \rfloor$. A method of alternately finding the best parameter combination is used. In detail, we first fix K to 5, and increase p by the step of 0.5 from the initial value 0.1. Until the performance of MIML-LLMC does not increase or even decrease due to the change of p , the approximate value range of p is obtained. The results show that the performance of the algorithm is acceptable when the value of p lies in $[0.1, 3]$. The same method is used to obtain the approximate value range of K, which is the interval $[5, 50]$ by the step of 5. For brevity, only the $Average\ Precision$ is employed here to show the results of our algorithm with different values of p and K. In order to more intuitively exhibit the performance gap, all the results of one dataset are first subtracted from their minimum value, and then normalized.

For moderate-classed datasets, as illustrated in Fig. 5a to 5e, the suitable values for K lie in the interval $[15, 30]$ and that of p is in $[0.5, 1.5]$. Furthermore, results with a too large K tend to be worse than those with a too small one. The reason may be that an excessively large neighborhood will introduce disturbing information to mislead the reconstruction of the label manifold.

For large-classed datasets, a problem will be encountered during the generation of the LLMCs. When we use only part of the bags for training, the generated numeric labels will be less than the labels. Since the number of labels is almost the same as that of bags. For example, in this paper, the ratio of training set is $2/3$, which means there are only 252 training numerical labels for GEO, but the number of labels is 320. It indicates that the range of values for p cannot be as large as that on moderate-classed datasets. Because a large p will cause the number of clustering centers in the label manifold to be more than the total number of numerical labels, the algorithm will not be able to work. Therefore, the value of ratio p for GEO and AZO can only be enumerated from $[0.1, 0.7]$, where 0.7 is the largest possible value via manually selection by the step of 0.1. For HAL and PYR, the value of p can reach 0.8 The range of K is the same as that of moderate-classed datasets. We can conclude from Fig. 5f to 5i, HAL and PYR prefer a large K while GEO and AZO do the opposite.

D. Ablation Study

If the local label correlation is ignored, the proposed algorithm degenerates into a simple parallel attention network. This variant is denoted by MIML-LLMC$_1$. If only one attention mechanism is used to predict the label vector of the bag, the proposed algorithm loses the ability to learn the instance-label relationship. This variant is denoted by MIML-LLMC$_2$. We perform the ablation experiments by comparing the results of MIML-LLMC and its two variants on 5 moderate-classed datasets. We can conclude from Table II that the performance of MIML-LLMC$_1$ is slightly better than that of MIML-LLMC$_2$. We assume it may be caused by the different amounts of parameters. Although MIML-LLMC$_1$ does not consider local label correlation, the network of it contains more learnable parameters than that of MIML-LLMC$_2$. Hence, MIML-LLMC$_1$ outperforms MIML-LLMC$_2$ a little even without giving a consideration to local label correlation. Furthermore, MIML-LLMC outperforms the two variants on all moderate-classed datasets, which validates its effectiveness.

TABLE II: $Hamming\ loss$ (mean$_{\pm\text{std}}$) of ablation experiments. The best results are bolded.

Datasets	MIML-LLMC$_1$	MIML-LLMC$_2$	MIML-LLMC
Scene	$.1831_{\pm.0052}$	$.1934_{\pm.0037}$	$\mathbf{.1636_{\pm.0015}}$
Reuters	$.0352_{\pm.0044}$	$.0392_{\pm.0022}$	$\mathbf{.0273_{\pm.0011}}$
MSRC v2	$.0783_{\pm.0023}$	$.0821_{\pm.0024}$	$\mathbf{.0674_{\pm.0021}}$
Letter Carroll	$.1343_{\pm.0012}$	$.1517_{\pm.0024}$	$\mathbf{.1223_{\pm.0043}}$
Letter Frost	$.1393_{\pm.0031}$	$.1476_{\pm.0062}$	$\mathbf{.1228_{\pm.0022}}$

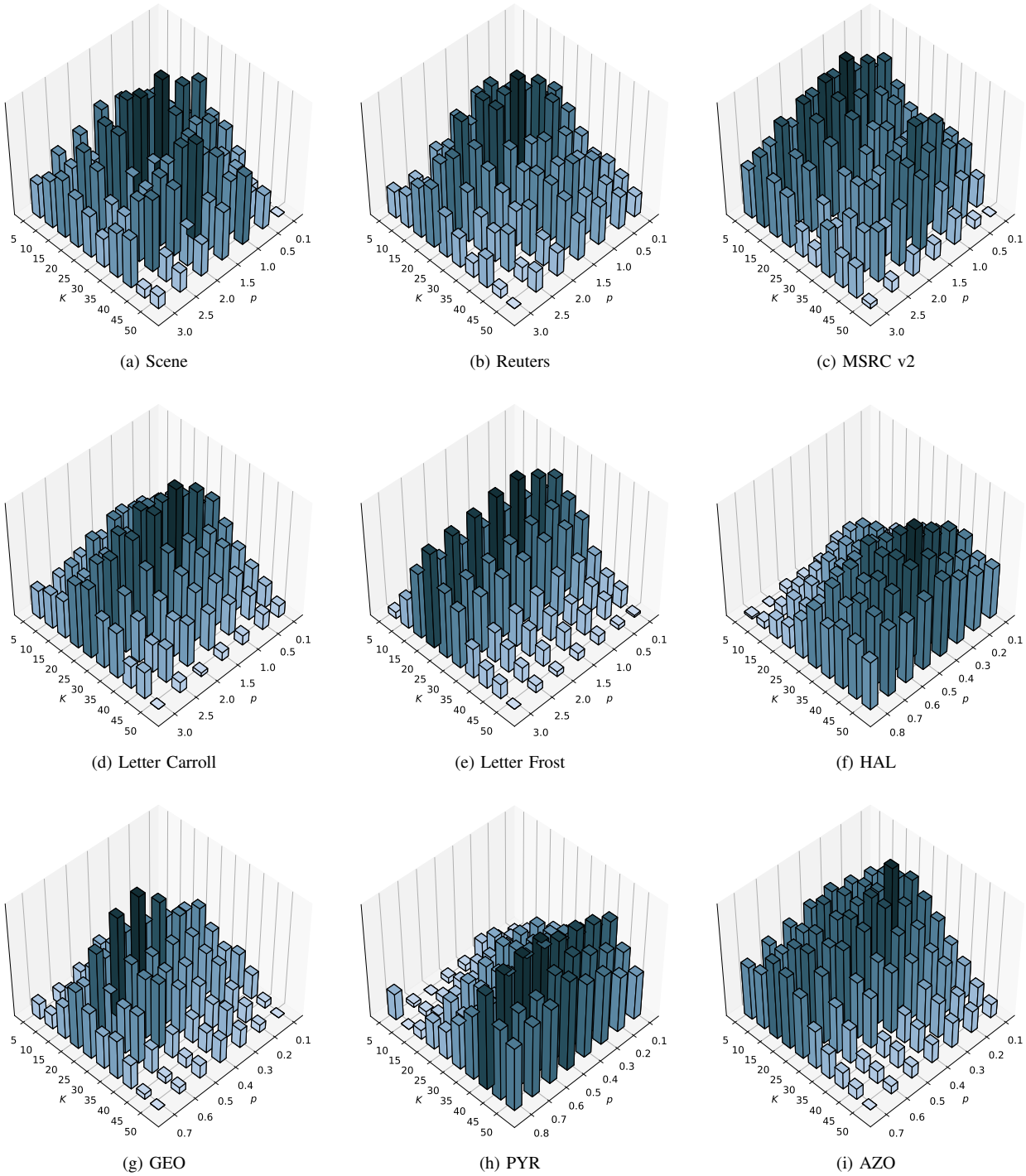
E. Comparison Algorithms

Our algorithm is compared with 6 state-of-the-art algorithms, which can also be divided into two groups. The former algorithms (MIMLSVM [4], MIMLRBF [9], MIMLKNN [11]) are developed using decomposition strategy, which break down the MIML tasks into a set of MIL or MLL tasks. The latter algorithms (KISAR [5], DeepMIML [6], MIMLFast [1]) are designed to tackle MIML tasks directly by special loss functions or networks.

MIMLSVM maps each bag into single vector based on the centroids obtained by performing k-Medoids on bag space. It further trains a multi-label SVM [39] using the mapped vectors. MIMLRBF partitions bags into L overlapping groups based on the labels they belong to, where L is the number of categories. Then the k-Medoids is also used to select bags from each group to construct a Gaussian based mapping function. MIMLKNN employs citers that regard the target bag as their own neighbors, to expand the traditional k-nearest algorithm. The label vector of each bag is determined by the label vectors of its neighbors and citers.

KISAR finds several prototypes of instances to map each bag into a single vector and formulates the task into a convex optimization problem. DeepMIML constructs a 3D sub-concept layer which can automatically locate the key input patterns that trigger the labels. MIMLFast designs a shared

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(a) Scene (b) Reuters (c) MSRC v2

(d) Letter Carroll (e) Letter Frost (f) HAL

(g) GEO (h) PYR (i) AZO

Fig. 5: Results comparison of MIML-LLMC on different datasets with different parameter settings. The results of GEO and AZO are under $One\ \ Error$ while others are under $Hamming\ \ Loss$.

sub-space for all labels and uses triples to optimize the a bag label ranking loss.

The parameters of above algorithms are set to the recommended values in the original papers.

F. Performance Comparison

As shown in Table III, our method achieves best results in 77.1% of cases and obtains the highest mean ranks under all 5 metrics. As aforementioned, each result is obtained by averaging the results of 10 times 3 CV. The series of these

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TABLE III: Comparison results (mean$_{\pm \text{std}}$) on moderate-classed datasets. ↑ (↓) indicates that the larger (smaller) the value, the better the performance. •(◦) indicates that MIML-LLMC is significantly better (worse) than the corresponding method based on paired t-tests at 95% significance level.

Datasets	MIMLSVM	MIMLRBF	MIMLKNN	KISAR	DeepMIML	MIMLFast	MIML-LLMC
				Hamming Loss ↓			
Scene	$.1812_{\pm.0047}$•	$.1942_{\pm.0112}$•	$.1931_{\pm.0022}$•	$.1841_{\pm.0021}$•	$.1714_{\pm.0102}$•	$.1691_{\pm.0038}$•	$.1636_{\pm.0015}$
Reuters	$.0693_{\pm.0053}$•	$.0923_{\pm.0133}$•	$.0809_{\pm.0014}$•	$.0553_{\pm.0015}$•	$.0326_{\pm.0014}$•	$.0343_{\pm.0012}$•	$.0273_{\pm.0011}$
MSRC v2	$.0784_{\pm.0042}$•	$.0777_{\pm.0012}$•	$.0802_{\pm.0013}$•	$.0804_{\pm.0006}$•	$.0693_{\pm.0041}$	$.0670_{\pm.0024}$	$.0674_{\pm.0021}$
Letter Carroll	$.1351_{\pm.0084}$•	$.1525_{\pm.0014}$•	$.1393_{\pm.0052}$•	$.1495_{\pm.0012}$•	$.1315_{\pm.0023}$•	$.1242_{\pm.0013}$•	$.1223_{\pm.0043}$
Letter Frost	$.1306_{\pm.0062}$•	$.1402_{\pm.0033}$•	$.1464_{\pm.0073}$•	$.1392_{\pm.0001}$•	$.1262_{\pm.0022}$•	$.1264_{\pm.0012}$•	$.1228_{\pm.0022}$
GEO	$.0106_{\pm.0007}$•	$.0108_{\pm.0001}$•	$.0159_{\pm.0023}$•	$.0101_{\pm.0003}$•	$.0103_{\pm.0008}$•	$.0099_{\pm.0008}$	$.0095_{\pm.0001}$
AZO	$.0115_{\pm.0016}$	$.0120_{\pm.0001}$•	$.0125_{\pm.0011}$•	$.0121_{\pm.0001}$•	$.0120_{\pm.0001}$•	$.0118_{\pm.0002}$	$.0117_{\pm.0001}$
Mean Rank	4.00	5.86	6.29	5.14	3.14	2.29	1.29
				One Error ↓			
Scene	$.3342_{\pm.0147}$•	$.3547_{\pm.0301}$•	$.3356_{\pm.0053}$•	$.3512_{\pm.0034}$•	$.2992_{\pm.0101}$•	$.2926_{\pm.0071}$•	$.2813_{\pm.0022}$
Reuters	$.1283_{\pm.0143}$•	$.1375_{\pm.0097}$•	$.1524_{\pm.0036}$•	$.1104_{\pm.0042}$•	$.0744_{\pm.0012}$•	$.0527_{\pm.0013}$•	$.0496_{\pm.0019}$
MSRC v2	$.2841_{\pm.0371}$•	$.2835_{\pm.0131}$•	$.3058_{\pm.0104}$•	$.2956_{\pm.0054}$•	$.2337_{\pm.0124}$•	$.2145_{\pm.0112}$	$.2061_{\pm.0081}$
Letter Carroll	$.3726_{\pm.0733}$•	$.5962_{\pm.0314}$•	$.3285_{\pm.0433}$•	$.3068_{\pm.0071}$•	$.2193_{\pm.0053}$	$.2202_{\pm.0122}$	$.2212_{\pm.0424}$
Letter Frost	$.4195_{\pm.0642}$•	$.6541_{\pm.0503}$•	$.4312_{\pm.0192}$•	$.4571_{\pm.0004}$•	$.3901_{\pm.0132}$•	$.3544_{\pm.0021}$•	$.3395_{\pm.0231}$
GEO	$.6516_{\pm.0359}$•	$.7398_{\pm.0074}$•	$.7937_{\pm.0141}$•	$.6594_{\pm.0039}$•	$.6341_{\pm.0099}$•	$.6216_{\pm.0049}$	$.6195_{\pm.0057}$
AZO	$.7042_{\pm.0290}$•	$.7636_{\pm.0027}$•	$.7552_{\pm.0118}$•	$.6992_{\pm.0090}$•	$.6793_{\pm.0041}$•	$.6692_{\pm.0130}$•	$.6515_{\pm.0064}$
Mean Rank	4.71	6.29	6.00	5.00	2.71	2.00	1.29
				Coverage ↓			
Scene	$.1994_{\pm.0072}$•	$.2091_{\pm.0241}$•	$.1931_{\pm.0012}$•	$.2062_{\pm.0012}$•	$.1842_{\pm.0042}$	$.1815_{\pm.0125}$	$.1781_{\pm.0063}$
Reuters	$.0553_{\pm.0056}$•	$.0712_{\pm.0044}$•	$.0832_{\pm.0015}$•	$.0546_{\pm.0012}$•	$.0224_{\pm.0026}$•	$.0142_{\pm.0022}$	$.0115_{\pm.0025}$
MSRC v2	$.2142_{\pm.0174}$	$.2414_{\pm.0077}$•	$.2455_{\pm.0047}$•	$.3161_{\pm.0092}$•	$.2192_{\pm.0043}$•	$.2224_{\pm.0091}$•	$.2112_{\pm.0055}$
Letter Carroll	$.4701_{\pm.0322}$•	$.4553_{\pm.0159}$•	$.4624_{\pm.0182}$•	$.4204_{\pm.0014}$	$.4096_{\pm.0132}$	$.4011_{\pm.0023}$◦	$.4112_{\pm.0202}$
Letter Frost	$.5186_{\pm.0321}$•	$.4672_{\pm.0132}$•	$.5312_{\pm.0173}$•	$.5682_{\pm.0011}$•	$.4553_{\pm.0122}$	$.4512_{\pm.0046}$•	$.4464_{\pm.0141}$
GEO	$.5897_{\pm.0200}$•	$.5807_{\pm.0137}$•	$.5957_{\pm.0141}$•	$.5467_{\pm.0037}$•	$.5288_{\pm.0107}$•	$.5153_{\pm.0132}$•	$.4901_{\pm.0138}$
AZO	$.5843_{\pm.0149}$•	$.5776_{\pm.0061}$•	$.5801_{\pm.0051}$•	$.5471_{\pm.0061}$•	$.5092_{\pm.0091}$•	$.5172_{\pm.0061}$•	$.4823_{\pm.0114}$
Mean Rank	5.29	5.29	6.00	5.14	2.71	2.29	1.29
				Ranking Loss ↓			
Scene	$.1802_{\pm.0082}$•	$.1932_{\pm.0192}$•	$.1751_{\pm.0013}$•	$.1891_{\pm.0012}$•	$.1594_{\pm.0012}$•	$.1668_{\pm.0112}$•	$.1532_{\pm.0051}$
Reuters	$.0373_{\pm.0045}$•	$.0443_{\pm.0041}$•	$.0662_{\pm.0021}$•	$.0352_{\pm.0013}$•	$.0201_{\pm.0022}$•	$.0216_{\pm.0021}$•	$.0169_{\pm.0032}$
MSRC v2	$.0901_{\pm.0133}$◦	$.1117_{\pm.0052}$•	$.1204_{\pm.0032}$•	$.1792_{\pm.0110}$•	$.0915_{\pm.0022}$◦	$.0939_{\pm.0012}$	$.0933_{\pm.0014}$
Letter Carroll	$.2095_{\pm.0221}$•	$.2346_{\pm.0081}$•	$.1962_{\pm.0102}$•	$.2612_{\pm.0022}$•	$.1773_{\pm.0094}$•	$.1748_{\pm.0222}$	$.1640_{\pm.0132}$
Letter Frost	$.2432_{\pm.0193}$•	$.2312_{\pm.0071}$•	$.2551_{\pm.0114}$•	$.2815_{\pm.0011}$•	$.2111_{\pm.0051}$•	$.2001_{\pm.0082}$	$.1942_{\pm.0081}$
GEO	$.4343_{\pm.0266}$•	$.4463_{\pm.0137}$•	$.4445_{\pm.0122}$•	$.4005_{\pm.0076}$•	$.3750_{\pm.0066}$•	$.3149_{\pm.0062}$•	$.2984_{\pm.0062}$
AZO	$.4496_{\pm.0168}$•	$.4715_{\pm.0135}$•	$.4401_{\pm.0066}$•	$.4009_{\pm.0010}$•	$.3704_{\pm.0036}$•	$.3401_{\pm.0066}$•	$.3032_{\pm.0078}$
Mean Rank	4.57	6.00	5.43	5.57	2.57	2.57	1.29
				Average Precision ↑			
Scene	$.7832_{\pm.0081}$•	$.7563_{\pm.0192}$•	$.7851_{\pm.0026}$•	$.7726_{\pm.0015}$•	$.8102_{\pm.0012}$•	$.7982_{\pm.0013}$•	$.8141_{\pm.0014}$
Reuters	$.9241_{\pm.0074}$•	$.8906_{\pm.0103}$•	$.8963_{\pm.0012}$•	$.9322_{\pm.0023}$•	$.9701_{\pm.0026}$	$.9640_{\pm.0023}$	$.9693_{\pm.0061}$
MSRC v2	$.7219_{\pm.0264}$•	$.7182_{\pm.0101}$•	$.6915_{\pm.0016}$•	$.6021_{\pm.0041}$•	$.7556_{\pm.0031}$•	$.7623_{\pm.0012}$	$.7614_{\pm.0032}$
Letter Carroll	$.5733_{\pm.0341}$•	$.4662_{\pm.0114}$•	$.5996_{\pm.0212}$•	$.5594_{\pm.0026}$•	$.6337_{\pm.0025}$•	$.6502_{\pm.0082}$	$.6534_{\pm.0021}$
Letter Frost	$.5116_{\pm.0192}$•	$.4275_{\pm.0142}$•	$.5039_{\pm.0122}$•	$.5312_{\pm.0012}$•	$.5719_{\pm.0132}$	$.5601_{\pm.0013}$•	$.5716_{\pm.0032}$
GEO	$.2768_{\pm.0225}$•	$.1821_{\pm.0092}$•	$.1771_{\pm.0076}$•	$.2732_{\pm.0086}$•	$.2952_{\pm.0106}$•	$.2932_{\pm.0006}$•	$.3039_{\pm.0046}$
AZO	$.2409_{\pm.0202}$•	$.1676_{\pm.0027}$•	$.1705_{\pm.0034}$•	$.2639_{\pm.0008}$•	$.2809_{\pm.0002}$•	$.2702_{\pm.0032}$•	$.2857_{\pm.0032}$
Mean Rank	4.71	6.57	5.57	5.14	2.00	2.57	1.43

10 results are used to explore whether there is a significant performance difference between the algorithms. If the p-value of two series is larger than 0.05 (95% significance level), the null-hypothesis that there is no significant difference between the respective algorithms' performance is rejected.

We first report the comparison results on the moderate-classed datasets in Table III. MIMLSVM tends to favor image data (Scene and MSRC v2) and performs better under *Hamming Loss* and *Ranking Loss*, while MIML-LLMC outperforms it on the rest datasets under all metrics. MIML-RBF and MIMLKNN work steadily on all datasets, but neither yields good results. The performance of KISAR is better on the text datasets (Reuters, Letter Carroll and Letter Frost) than on the image datasets, but still not satisfactory. DeepMIML and MIMLFast achieve good results but are outperformed by MIML-LLMC in most cases, especially on Scene and Reuters. Among all 6 comparison algorithms on 5 datasets, only the results of MIMLSVM and DeepMIML under *Ranking Loss*

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Fig. 6: The label of each instance (region) obtained by MIML-LLMC. Region corresponding to each label is highlighted in red.

are significant better than that of ours.

On large-classed datasets, due to the large number of labels, most comparison algorithms cannot obtain acceptable results, especially under the last 4 metrics. However, our algorithm maintains best results in 9 out of 10 cases and the comparison algorithms are significant worse in most cases. MIMLSVM gets the best results on AZO under $Hamming\ Loss$, but the difference to MIML-LLMC is not significant. MIMLRBF, MIMLKNN, and KISAR obtain stable but average performance. DeepMIML and MIMLFast rank top 3 on both datasets under most metrics. In detail, DeepMIML is significantly worse that MIML-LLMC in all cases, whereas MIMLFast is also significantly worse in 7 out of 10 cases.

G. Instance Label Accuracy

In MIML, one critical thing is to find out which instance triggers which label. According to Eq. (12), a label triggered by an instance is identified as the instance's label. In this way, MIML-LLMC is able to learn instance labels using only bag labels. Following the method in [1], we first provide a intuitive display of instance labels using the original images in MSRC v2 [2]. Each image is segmented into several regions, each region is extracted as an instance and pixel-wise labelled by the annotators. As shown in Fig. 6, our approach provides reasonable results on understanding the relation between instances and labels.

Some experiments are conducted to evaluate the performance quantitatively. The instance labels are available in MSRC v2, Letter Carroll and Letter Frost, which offers a test bed to obtain instance label accuracy. In aforementioned comparison algorithms, KISAR, DeepMIML, and MIMLFast can also learn instance labels with only bag labels. KISAR treats the label of the nearest prototype as the label of the instance. The instance labels in DeepMIML are obtained by performing max-pooling along the row of the 3D-concept layer. MIMLFast learns the label for one instance by selecting which label has the maximum prediction value of the instance. To provide a benchmark performance for comparison, we design a single-instance multi-label toy algorithm which utilizes multiple linear SVMs to predict each label individually. Furthermore, according to Eq. (14), we propose a variant of our algorithm called MIML-LLMC$_{ins}$ to exploit instance-level supervision. MSE is utilized as the bag-level loss function $\mathcal{L}^B()$, and the instance-level loss function $\mathcal{L}^I()$ is cross-entropy. The trading-off parameter λ is enumerated from $\{1, 2, \ldots, 10\}$. The results of instance label prediction are shown in Table IV. As expected, toy algorithm outperforms all algorithms without instance-level supervision and only MIMLFast and MIML-LLMC can offer similar results to it. By directly connecting attention mechanisms to the instance-level loss function $\mathcal{L}^I()$, MIML-LLMC$_{ins}$ exhibits the best performance unsurprisingly.

TABLE IV: $Accuracy$ (mean$_{\pm\text{std}}$) of instance label prediction. The best results are bolded.

Algorithms	MSRC v2	Letter Carroll	Letter Frost
KISAR	$0.17_{\pm 0.06}$	$0.23_{\pm 0.02}$	$0.29_{\pm 0.03}$
DeepMIML	$0.19_{\pm 0.02}$	$0.21_{\pm 0.08}$	$0.35_{\pm 0.01}$
MIMLFast	$0.27_{\pm 0.04}$	$0.38_{\pm 0.02}$	$0.36_{\pm 0.01}$
MIML-LLMC	$0.26_{\pm 0.04}$	$0.40_{\pm 0.06}$	$0.35_{\pm 0.03}$
Toy	$0.33_{\pm 0.02}$	$0.41_{\pm 0.05}$	$0.40_{\pm 0.05}$
MIML-LLMC$_{ins}$	$\mathbf{0.56_{\pm 0.07}}$	$\mathbf{0.62_{\pm 0.03}}$	$\mathbf{0.62_{\pm 0.05}}$

[2] http://research.microsoft.com/en-us/projects/objectclassrecognition/default.htm

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V. Conclusion

As a natural extension of traditional supervised learning, MIML has been proven to be effective in many applications. However, the existing algorithms mainly focus on handling bag structure and understanding the relation between instances and labels, while the label space is ignored. To address this issue, we propose the MIML-LLMC algorithm to exploit the local label correlation. On the one hand, with label manifold, the local label correlation is encoded as an additional vector (LLMC) for each bag and used for prediction. On the other hand, the parallel attention mechanisms and two level loss function enable the model to discover the relation between input patterns and output label semantics. In the future, we will try other method to exploit more abundant information from label space. Moreover, the relationship between instances and labels will be further studied.

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